

KuMMLU: A Multitask Language Understanding Benchmark for Central Kurdish

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Abstract

Language models are now tested across many languages, but Kurdish is mostly missing from these efforts. We present KuMMLU, a multiple-choice benchmark for Central Kurdish (Sorani) built from the national examinations of the Kurdistan Region of Iraq. It contains 3,272 questions across 18 subjects: 2,256 native examination questions, with answers taken from the official keys published by the Ministry of Education, and 1,016 questions from six subjects translated from English MMLU. We evaluate 32 language models, including proprietary systems, open-weight models, and multilingual models built for wide language coverage. The best models reach about 88% accuracy, while the five multilingual baselines score between 15.8% and 37.2%, at most modestly above the 25% chance level and well below the strongest general-purpose systems; this includes the one model whose published language list contains Kurdish. Every model scores higher on the translated subset than on the native questions. We examine this gap in light of subject difficulty and possible overlap with training data, and release the benchmark to support Kurdish language technology.

1 Introduction

Kurdish is spoken by tens of millions of people, yet it remains under-served by language technology (Esmaili, 2012; Ahmadi et al., 2024). It has three main varieties, Northern Kurdish (Kurmanji), Central Kurdish (Sorani), and Southern Kurdish, which differ in vocabulary, morphology, and writing system (Haig and Öpengin, 2014). Sorani is written in a modified Perso-Arabic script and holds official status in Iraq (Ahmadi et al., 2024). Most tools and datasets for Kurdish target basic tasks such as tokenization, transliteration, part-of-speech tagging, or named entity recognition (Ahmadi, 2020; Morad et al., 2024; Abdullah et al., 2024), and there is no standard way to measure what a modern language

model knows in Kurdish.

Multiple-choice benchmarks based on MMLU (Hendrycks et al., 2021) have become a common tool for this kind of measurement. Versions now exist for Arabic (Koto et al., 2024), Turkish (Yüksel et al., 2024), Korean (Son et al., 2025), and many languages at once (Singh et al., 2025), but not for Kurdish. Translated benchmarks are also known to reward Western-centric knowledge and to carry translation artifacts (Singh et al., 2025), so native questions are preferable when they are available.

We address this gap with KuMMLU, a benchmark for Central Kurdish drawn from national examinations. Our contributions are:

- A benchmark of 3,272 Sorani multiple-choice questions across 18 subjects, most of them written natively in Kurdish and answered from official examination keys.
- An evaluation of 32 language models, covering proprietary, open-weight, and multilingual systems.
- An analysis of the gap between native and translated questions, and of how multilingual models built for wide language coverage perform on Kurdish.

2 Related Work

MMLU and its variants. MMLU (Hendrycks et al., 2021) measures knowledge and reasoning through multiple-choice questions in 57 subjects. Later work adapts the format to specific languages. Some versions translate the English questions, while others collect native questions from local exams. KMMLU (Son et al., 2025) and ArabicMMLU (Koto et al., 2024) both argue that native questions capture linguistic and cultural content that translation loses, and TurkishMMLU (Yüksel et al., 2024) builds its set from high-school curricula. Global-MMLU (Singh et al., 2025) studies cul-

tural bias across languages and improves the translated MMLU with human annotation. KuMMLU follows the native-collection approach and adds a translated subset for comparison.

Kurdish NLP. Work on Kurdish has focused on foundational resources and tools. Esmaili (2012) describes the main obstacles, including dialect diversity and script variation. The Kurdish Language Processing Toolkit (Ahmadi, 2020) provides pre-processing and transliteration, and Ahmadi et al. (2024) survey language and speech technology for Central Kurdish. Recent task-specific systems include part-of-speech tagging for Northern Kurdish (Morad et al., 2024) and named entity recognition for Sorani built by fine-tuning transformer encoders (Abdullah et al., 2024). To our knowledge, no prior benchmark measures the general knowledge of language models in Kurdish.

Multilingual models. Several models are trained for broad language coverage. mT0 and BLOOMZ (Muennighoff et al., 2023) extend multitask fine-tuning to many languages, and the Aya models (Üstün et al., 2024; Dang et al., 2024) target instruction following across dozens to over a hundred languages. We include these models in our evaluation because they are the systems most likely to have seen Kurdish during training.

3 The KuMMLU Benchmark

3.1 Sources

The native questions come from the national examinations of the Kurdistan Region of Iraq, at the Grade 9 and Grade 12 levels. These are the two grade levels at which the Kurdistan Region holds centralized national examinations; they are set by the KRG Ministry of Education and cover the standard school curriculum. We recorded the correct answers manually from the official answer keys published by the Ministry, so each answer reflects the position taken by the examining authority and was not decided by us. A second person then checked the recorded answers against the published keys to catch transcription errors.

To connect KuMMLU to the original MMLU, we added a translated subset of six subjects: high school mathematics, philosophy, college chemistry, college physics, college mathematics, and college medicine. We translated these from English MMLU (Hendrycks et al., 2021) with Gemini 2.5 Flash and kept the original MMLU answer labels.

Two native Sorani speakers reviewed the translated questions and corrected fluency and terminology errors. This subset lets us compare native and translated questions inside one benchmark, with the caveats discussed in Limitations.

3.2 Annotation and quality control

The exam material was collected as page images and passed through optical character recognition. Two native Sorani speakers then checked the questions against their source images and corrected recognition errors, spacing, and spelling. Each question was checked by one speaker, and a second speaker reviewed the flagged items along with a random sample of the rest. We wrote short guidelines that told the checkers to keep the original wording, to leave technical terms and formulas unchanged, and to flag any item whose text or answer could not be recovered from the source. Cases that were unclear were resolved by discussion between the two speakers.

We then removed duplicate questions that appeared more than once in the source material, and discarded items with fewer than two usable options. We also normalized Arabic-script code points that OCR had introduced in place of their Kurdish equivalents, while keeping genuine Arabic quotations such as Qur’anic verses unchanged. As a further check, we used Gemini 3.1 Pro to flag questions with likely spelling or OCR problems, and gave those flags to the native speakers, who reviewed each flagged item and decided whether to correct it.

3.3 Statistics

The final benchmark has 3,272 questions in 18 subjects: 2,256 questions (12 subjects) written natively in Kurdish, and 1,016 (6 subjects) translated from English MMLU. The per-subject counts are in Appendix A. Of the 3,272 questions, 3,264 have four options; seven Kurdish literature questions have three and one mathematics question has five, as in the source papers. After the option shuffling described in Section 4.1, the correct answer falls in each position for 24.7–25.6% of the four-option questions.

4 Experiments

4.1 Setup

We evaluate each model in a zero-shot setting, which is the common protocol for instruction-tuned

Model	T	All	Nat	Tr	Model	T	All	Nat	Tr
Gemini 3.1 Pro	P	88.1	85.5	93.6	GPT-4o	P	46.1	44.2	50.3
Gemini 3.6 Flash	P	87.8	84.6	95.1	Qwen3 235B [‡]	O	45.6	42.0	53.5
GPT-5.6 (sol)	P	82.9	78.0	93.8	Llama 4 Scout [‡]	O	44.4	42.9	47.8
Gemini 2.5 Pro	P	75.9	73.4	81.7	Llama 3.3 70B [‡]	O	41.6	39.1	47.3
Claude Opus 4.8	P	75.4	70.5	86.2	Mistral Large	O	38.4	36.1	43.7
Kimi K3 [‡]	O	72.6	66.7	85.6	Aya Expanse 32B	M	37.2	35.8	40.2
Qwen3.7 Max	P	69.5	65.8	77.9	Mistral Medium 3.1	P	36.2	34.2	40.6
Claude Sonnet 5	P	68.7	62.4	82.6	Qwen3 8B	O	33.9	32.6	36.8
GPT-5.6 (luna) [‡]	P	68.5	60.5	86.3	Mistral Small 24B	O	32.1	30.5	35.7
Gemini 2.5 Flash	P	68.1	67.3	69.8	Aya Expanse 8B	M	32.1	31.2	34.0
Claude Sonnet 4	P	64.9	58.8	78.4	Ministral 8B	O	31.4	30.2	34.3
Gemini 2.5 Flash-Lite	P	62.3	59.9	67.7	Ministral 14B	O	31.3	30.0	34.2
Gemma 4 31B	O	57.9	53.6	67.2	Aya-101	M	29.5	29.0	30.5
Llama 4 Maverick	O	57.5	54.8	63.6	mT0-XXL [‡]	M	29.4	28.7	30.8
DeepSeek V4 Flash	O	53.8	51.7	58.4	Ministral 3B	O	27.5	27.2	28.3
Command A	O	48.3	45.0	55.6	BLOOMZ-7B1 [†]	M	15.8	14.5	18.8

Table 1: Accuracy (%) on KuMMLU, ordered by overall accuracy. Type: P = proprietary, O = open-weight, M = multilingual baseline. **All**, **Nat**, and **Tr** are overall, native-subset, and translated-subset accuracy. The chance level is approximately 25%. GPT-5.6 (sol) and GPT-5.6 (luna) are two configurations of GPT-5.6 offered by the provider. [†]BLOOMZ-7B1 left 37% of items unparsed; see text. [‡]Response rate between 83% and 98%; see Appendix F.

models on MMLU-style benchmarks. Each question is presented with its options, and the model is asked to reply with the option number. The instruction is written in English and the question and its options are kept in Kurdish; the exact template is given in Appendix B. We read the chosen option from the model’s output. Because language models are known to favour particular option positions (Zheng et al., 2024), we shuffle the options for each question with a fixed seed before evaluation and remap the correct answer accordingly. Reasoning (“thinking”) is set to low effort for GPT-5.6 (sol and luna), Claude Opus 4.8, Claude Sonnet 5, Claude Sonnet 4, Kimi K3, and the Gemini 2.5 models, and left at the provider default for Gemini 3.1 Pro and Gemini 3.6 Flash; for all other models it is disabled. We do not tune inference settings per model. API requests use temperature 0.

Proprietary and open-weight chat models are queried through the OpenRouter API; the multilingual baselines (mT0-XXL, BLOOMZ-7B1, and the Aya models) are run locally in bf16 on a single A100 GPU with the same prompt and shuffling. Accuracy counts an unparsed answer as incorrect. Response rates were above 98% for 25 of the 32 models and above 90% for all but two, BLOOMZ-7B1 (63%) and Qwen3 235B (83%). Per-model response rates are given in Appendix F, which lets missing or unparsable responses be separated from wrong answers. For the API-served models, most unparsed responses were empty replies.

4.2 Models

We test 32 models in three groups. The first is proprietary chat models from several providers. The second is open-weight models across a range of sizes. The third is multilingual models built for wide language coverage: mT0-XXL and BLOOMZ-7B1 (Muennighoff et al., 2023), Aya-101 (Üstün et al., 2024), and Aya Expanse in the 8B and 32B sizes (Dang et al., 2024). Aya-101 is the only model in our set whose published language list contains Kurdish, though it lists the Latin-script (Kurmanji) form, not the Arabic-script Sorani used here.

4.3 Results

Table 1 reports overall accuracy together with accuracy on the native and translated subsets. The strongest models reach about 88%. Accuracy then falls off across the open-weight models, and the multilingual baselines fall in the lower part of the table, at most modestly above the chance level (approximately 25%, since most questions have four options).

Two patterns hold across the whole table. First, every model scores higher on the translated subset than on the native questions. The six translated subjects come from English MMLU and overlap with material these models have likely seen in English, so this gap may reflect knowledge acquired elsewhere rather than Kurdish knowledge itself. The size of the gap is not uniform. It is smallest for the weakest models, which perform near chance

on both subsets (1.1 points for Ministral 3B), and largest among models in the middle and upper-middle of the table (up to 25.8 points for GPT-5.6 (luna)). Among the strongest models the picture is mixed: Gemini 3.1 Pro shows a gap of 8.1 points, while GPT-5.6 (sol) and Claude Opus 4.8 show gaps of 15.8 and 15.7 points. The Gemini models, whose family produced the translated subset, show smaller gaps than non-Gemini models of similar overall accuracy, but this comparison cannot isolate a possible advantage from Gemini-generated translations.

Second, the multilingual models built for many languages do poorly. mT0-XXL and Aya-101 score 29.4% and 29.5%, close to chance, and Aya Expanse 32B reaches only 37.2% despite being a newer and larger model. Aya-101 lists Kurdish among its languages but does not outperform Aya Expanse, which does not. This indicates that a language appearing on a training list does not by itself produce usable ability in that language. BLOOMZ-7B1 scores 15.8%, below chance: it failed to produce a parsable answer on 37% of items, and on the items it did answer it scored 25%, at chance. Its low aggregate therefore reflects both frequent failures to follow the answer format on Kurdish-language inputs and chance-level accuracy on the answers it gave.

Within the open-weight groups, larger models in a family tend to score higher, but even the best open model trails the strongest proprietary systems by a wide margin. One open model, Qwen3 235B, returned a parsable answer on only 83% of items, mostly because its replies were empty, and its accuracy on those items (54.7%) is well above its 45.6% overall score. Accuracy also varies by subject. Averaged across all models, the three Kurdish-language subjects are the hardest: rhetoric (33.7%), grammar (36.1%), and literature (40.8%) are the lowest of the 18 subjects, while general-knowledge subjects such as religious studies and social sciences are the highest. Full per-subject figures are in Appendix C. Subject sizes vary (Table 2), so estimates for the smallest subjects are less precise: rhetoric has only 49 questions, and the two highest-scoring subjects, social sciences and religious studies, have 55 and 68. The low scores on the Kurdish-language subjects do not rest on rhetoric alone, however, since grammar (282 questions) and literature (289) are among the largest subjects in the benchmark. This pattern fits the earlier observation: the subjects that are most specific

to Kurdish, and least likely to have a counterpart in higher-resource languages, are where models struggle most. Appendix D shows example questions, and Appendix E examines the errors in more detail.

5 Conclusion

We introduced KuMMLU, a multiple-choice benchmark for Central Kurdish built from national examinations, and used it to evaluate 32 language models. The best proprietary models reach about 88%, while multilingual models built for wide coverage score at most modestly above chance, and every model does better on translated questions than on native ones. These results give a first clear picture of where language models stand on Kurdish and provide a reference point for future work. We plan to extend KuMMLU to Kurmanji, so that a second Kurdish variety is covered.

Limitations

The benchmark covers Central Kurdish in the Arabic script only; Kurmanji and the Latin script are left for future work. The questions come from national school examinations, so the benchmark reflects that curriculum and not the full range of Kurdish knowledge.

The comparison between the native and translated subsets should be read with care, for three reasons. First, the two subsets differ in difficulty and domain: the native questions are from Grade 9 and Grade 12 school exams, while the translated questions come from MMLU’s high-school and college-level subjects. Part of the native–translated gap may therefore reflect difficulty, not language. Second, the translated subjects are taken from English MMLU, which is widely available online and is likely present in the training data of many models (Oren et al., 2024), so high scores there may reflect memorized English questions rather than Kurdish ability. The native exam papers also circulate online and could appear in training data, though we are not aware of a released, machine-readable version of them. Third, the translated questions were produced with a Gemini model, and Gemini models are among those we evaluate. The per-model gaps in Table 1 do not suggest that this gives them an advantage on the translated subset, but they cannot rule it out. We therefore report the gap as an observation and do not claim a single cause.

On the evaluation itself: we shuffle options once with a fixed seed, which reduces but does not re-

move position effects; averaging over several permutations, or scoring by option likelihood, would be more thorough, and the latter would also measure format-sensitive models such as BLOOMZ more fairly. The open-weight models are served through an API aggregator and may be quantized, so their scores may not exactly match the full-precision weights. The proprietary models are queried at a fixed point in time and may change. We evaluated each model once, using the inference settings described in Section 4.1 without performance-based tuning, so results may differ under other settings such as higher reasoning effort or few-shot prompting. The instruction is written in English; a Kurdish instruction could change the results, particularly for weaker models that already struggle with the answer format. Finally, the translated subset was checked by native speakers but may still contain translation errors. We did not formally measure inter-annotator agreement during checking; unclear cases were resolved by discussion, as described in Section 3.2.

Ethics Statement

The native questions come from national examination papers and their official answer keys, which are public documents, and the translated questions come from the publicly released MMLU benchmark. It contains no personal or private information. Two native Sorani speakers took part in the checking process; they were university students who volunteered for the project and were informed of its purpose and intended use. We release the benchmark, the evaluation code, and the per-model results at <https://huggingface.co/datasets/akam-ot/KuMMLU> under the CC BY-NC 4.0 license for research use, with documentation of the sources, the annotation process, and the native/translated split. Because publicly released benchmarks can be absorbed into future training data, the release includes a canary string and asks that the data not be used for training, so that the benchmark stays useful for evaluation.

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A Subjects

Table 2 lists the subjects in KuMMLU and the number of questions in each.

Subject	#Q
<i>Native (2,256)</i>	
Chemistry	336
Kurdish literature	289
Kurdish grammar	282
Biology	259
History	256
Economics	178
Geography	166
Physics	161
Mathematics (Gr. 9 and 12)	157
Religious studies	68
Social sciences	55
Kurdish rhetoric	49
<i>Translated from MMLU (1,016)</i>	
Philosophy	311
High school mathematics	264
College medicine	164
College chemistry	99
College mathematics	99
College physics	79
Total	3,272

Table 2: Subjects in KuMMLU and their question counts.

B Prompt Template

Each question is presented with the template below, shown here for a four-option question. The instruction is in English and the question and options are kept in Kurdish. For the API-served models the first line is sent as the system message and the rest as the user message; for the locally run multilingual baselines the whole template is given as a single prompt. The number of options (N) is three, four, or five depending on the question, and the final line asks for a number from 0 to N−1.

You are taking a multiple choice exam in Kurdish. Respond with ONLY the number of the correct answer. No explanation.

Question: {question}

Choices:

0: {option 0}

1: {option 1}

2: {option 2}

3: {option 3}

Answer with a single number from 0 to {N−1}:

C Per-Subject Accuracy

Table 3 reports accuracy for each subject, averaged over the 32 models. The three Kurdish-language subjects are the lowest-scoring.

Subject	Mean acc. (%)
Kurdish rhetoric	33.7
Kurdish grammar	36.1
Kurdish literature	40.8
Geography	44.6
History	45.4
College chemistry	48.0
Economics	49.1
Biology	49.7
College mathematics	53.6
High school mathematics	56.7
Chemistry	57.1
Physics	58.4
College physics	59.7
Philosophy	59.9
Mathematics (Gr. 9 and 12)	61.6
College medicine	63.2
Social sciences	63.4
Religious studies	65.3

Table 3: Per-subject accuracy, averaged over all 32 models.

D Example Questions

Table 4 gives English glosses of four native questions, one from each of grammar, rhetoric, history, and chemistry. The correct option follows the official answer key. The last column gives how many of the 27 API-served models answered correctly.

E Error Analysis

This analysis uses the per-question outputs of the 27 models served through the API; the multilingual baselines are not included.

Where the Kurdish-language subjects lose points. The drop on the Kurdish-language subjects is not limited to weak models. The five strongest models lose 16 to 30 points on grammar, literature, and rhetoric compared with the other native subjects (Gemini 3.1 Pro: 74.2% vs. 89.9%; GPT-5.6 (sol): 56.5% vs. 86.2%). Of the 620 questions in these three subjects, 70 (11.3%) are answered correctly by at most three of the 27 models, compared with 4.0% of the other native questions and 2.7% of the translated ones.

Reading these hardest items shows three recurring types. Grammar questions often ask for the role of a Sorani attached pronoun (for example, whether *-m* is the agent or a prepositional complement) or for a rewrite that moves a clitic, which requires a correct analysis of the Sorani clitic system. Literature questions ask who wrote a quoted verse, or which statement matches the curriculum’s account of a literary movement or journal; here

models tend to choose a generally plausible statement or “all of the above” instead of the answer the curriculum teaches, knowledge that is likely to be scarce in higher-resource-language training data. Rhetoric questions ask models to identify or count the rhetorical devices in a verse.

Errors are shared. Wrong answers are concentrated. On questions that at least five of the ten strongest models answer incorrectly (counting parsable answers only), on average 73% of their wrong answers on the Kurdish-language subjects, and 79% on the other subjects, go to the same distractor. This is consistent with shared distractor preferences, although it does not by itself establish their cause.

Position bias. Option shuffling places the correct answer in each position for about a quarter of the questions (Section 4.1), and the strongest models’ predictions are also close to uniform: Gemini 3.1 Pro, GPT-5.6 (sol), and Kimi K3 choose each position between 23.7% and 26.8% of the time. Several open-weight models show strong preferences instead: Ministral 14B chooses the first option in only 4.0% of four-option questions and the third in 52.0%, and Llama 3.3 70B chooses the third option 46.8% of the time. Because the shuffle makes these preferences independent of the correct answer, they should not systematically inflate scores, although a single fixed ordering cannot rule out chance effects in either direction.

F Response Rates

Most models returned a parsable answer on nearly every question. Table 5 lists the models whose response rate fell below 98%; all other models were above 98%.

Model	Response rate (%)
BLOOMZ-7B1	62.9
Qwen3 235B	83.3
GPT-5.6 (luna)	91.7
Llama 4 Scout	93.8
Llama 3.3 70B	94.0
Kimi K3	95.9
mT0-XXL	96.2
<i>All other models</i>	>98

Table 5: Models with response rate below 98%. A low rate means the model often did not return a parsable option number.

Subject	Question and options (English gloss)	Models correct
Grammar	In the sentence <i>kam qutabî be yekem derçû?</i> ('Which student came first?'), what is <i>kam</i> ? (a) interrogative pronoun; (b) interrogative adverb; (c) interrogative adjective ; (d) adverb of manner	13/27
Rhetoric	In the verse ' <i>Ey biraderan, ba biçîne yarî / Newcewanîman lê bida diyarî</i> ', the imperfect paronomasia (<i>yarî/diyarî</i>) lies in: (a) the kind of letters; (b) the number of letters ; (c) the order of letters; (d) the kind and number of letters	10/27
History	The forces of Sheikh Ubeydullah of Nehri, for liberating the part of Kurdistan under Qajar rule, were divided into how many columns? (a) three ; (b) four; (c) five; (d) six	13/27
Chemistry	In which cell does an electric current drive a non-spontaneous redox reaction? (a) fuel cell; (b) voltaic cell; (c) electrolytic cell ; (d) dry cell	17/27

Table 4: Example native KuMMLU questions with English glosses. Bold marks the answer in the official key.